Math 241, F04 **Name:**
Quiz 3 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (10 points) Let $\mathbf{a} =< 3, 3, 1 >$, $\mathbf{b} =< -2, -1, 1 >$ and $\mathbf{c} =< -4, -3, -1 >$, find each of the following:

a) $\mathbf{a} \times \mathbf{b}$

Solution:

$$\begin{aligned}
\mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 3 & 1 \\ -2 & -1 & 1 \end{vmatrix} \\
&= \begin{vmatrix} 3 & 1 \\ -1 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 3 & 1 \\ -2 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 3 & 3 \\ -2 & -1 \end{vmatrix} \mathbf{k} \\
&= 4\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}
\end{aligned}$$

b) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

Solution: Since

$$\mathbf{a} \cdot \mathbf{b} =< 3, 3, 1 > \cdot < -2, -1, 1 >= -8$$
$$\mathbf{a} \cdot \mathbf{c} =< 3, 3, 1 > \cdot < -4, -3, -1 >= -22$$

We have

$$\begin{aligned}
\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} \\
&= -22 < -2, -1, 1 > -(-8) < -4, -3, -1 > \\
&= < 44, 22, -22 > - < 32, 24, 8 > \\
&= < 12, -2, -30 >
\end{aligned}$$

Problem 2. (10 points) Find the symmetric equations of the line of intersection of the given pair of planes:

$$x - 3y + z = -1, \quad 6x - 5y + 4z = 9.$$

Solution: Since $\mathbf{u} =< 1, -3, 1 >$ is the normal to the first plane and $\mathbf{v} =< 6, -5, 4 >$ is the normal to the second plane, we know a vector parallel to the line of intersection can be given by:

$$\mathbf{w} = \mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -3 & 1 \\ 6 & -5 & 4 \end{vmatrix} = -7\mathbf{i} + 2\mathbf{j} + 13\mathbf{k}$$

Then we need to find a point in the line. Let $x = 0$, we have

$$\begin{aligned} -3y + z &= -1 \\ -5y + 4z &= 9 \end{aligned}$$

By solving it we get $y = 13/7, z = 32/7$, so we know (0,13/7,32/7) is in the line.

The symmetric equations are then given by

$$\frac{x}{-7} = \frac{y - 13/7}{2} = \frac{z - 32/7}{13}$$